front matter

preface

This book is part of a dream to contribute to the development of the field I am most passionate about—computer science, and particularly, software development. These fields show their extraordinary importance in the interconnected and global present. We see the incredible transformations of these disciplines every day, in all areas of human activities. But why write about microservices architecture when there are a lot of other topics to write about?

The word “microservices” has a lot of interpretations. But in this book, I define a microservice as a distributed, loosely coupled software service that carries out a small number of well-defined tasks. Microservices crept up as an alternative to monolithic applications to help combat the traditional problems of complexity in a large code base by decomposing that down into small, well-defined pieces.

During my 13 years of experience, I have dedicated myself to software development, working with different languages and different types of software architectures. When I started, I used architectures that today are practically obsolete. The contemporary world forces us to update continually, and innovation in the software development field advances at an accelerated pace. Because of this search for up-to-date knowledge and practices, I decided to get involved in the world of microservices some years ago. Since then, it is the architecture I have used the most because of the advantages it provides (advantages such as scalability, speed, and maintainability). Successfully venturing into the microservices field prompted me to take on the task of writing this book as an opportunity to systematize and share what I have learned.

As a software developer, I discovered how important it is to continually research and apply new knowledge to development. Before undertaking this book, I decided to share my findings and started publishing microservices articles on the blog platform of a software development company I worked for in Costa Rica, my home country. While I was writing these articles, I realized I had found a new passion and purpose in my professional career. A few months after writing one of my articles, I received an email from Manning

Publications, offering me the opportunity to write the second edition of this book that I share with you today.

The first edition of this book was written by John Carnell, a consummate professional with many years of software development experience. I wrote this second edition from this basis, combined with my own interpretation and understanding. The second edition of *Spring Microservices in Action* will allow you to implement diverse design patterns that will help you create a successful microservices architecture using Spring. It's a framework that offers out-of-the-box solutions for many of the common development problems you will run into as a microservice developer. Now, let's start this incredible journey into the world of microservices with Spring.

acknowledgments

I am deeply grateful for the opportunity to work on this book, which has allowed me to share my knowledge and to learn at the same time. I am also grateful to Manning Publications for trusting my work and allowing me to share it with so many people; most importantly, to Michael Stephens for offering me this fantastic opportunity; to John Carnell for his support, work, and knowledge; to Robert Wenner, my technical development editor, for his valuable contributions; and to Lesley Trites, my editor, for accompanying me throughout the process with her valuable help.

I would also like to thank Stephan Pirnbaum and John Guthrie, who, as my technical reviewers, checked my work and ensured the book's overall quality. Thanks go also to my project editor, Deirdre Hiam; my copyeditor, Frances Buran; my proofreader, Katie Tennant; my reviewing editor, Aleks Dragosavljevic; and to all the reviewers (Aditya Kumar, Al Pezewski, Alex Lucas, Arpit Khandelwal, Bonnie Malec, Christopher Kardell, David Morgan, Gilberto Taccari, Harinath Kuntamukkala, Iain Campbell, Kapil Dev S, Konstantin Eremin, Krzysztof Kamyczek, Marko Umek, Matthew Greene, Philippe Vialatte, Pierre-Michel Ansel, Ronald Borman, Satej Kumar Sahu, Stephan Pirnbaum, Tan Wee, Todd Cook, and Víctor Durán)—your suggestions helped make this a better book.

I want to thank my mom, dad, and my entire family, who supported me and encouraged me to pursue my dreams, and who, by their dedication to work as an example, helped me become the professional I am today. My gratitude also goes out to Eli, who was always by my side on those long days of work, and to my friends, who always trusted me and encouraged me throughout this process.

Last, but not least, I thank each and every one of you for buying this book and allowing me to share my knowledge with you. I hope you enjoy reading it as much as I did writing

it. I hope it becomes a valuable contribution to your professional careers.

about this book

Spring Microservices in Action, Second Edition, is written for the practicing Java/Spring developer who needs hands-on advice and examples of how to build and operationalize microservice-based applications. When we wrote this book, we wanted to maintain the same central idea as in the first edition. We wanted it to be based on core microservice patterns aligned with Spring Boot and Spring Cloud's latest practices and examples. In almost every chapter, you will find specific microservice design patterns, along with examples of the Spring Cloud implementations.

Who should read this book

- You are a Java developer who has some experience (1–3 years) with building distributed applications.
- You have some background (1+ years) with Spring.
- You are interested in learning how to build microservice-based applications.
- You are interested in how you can leverage microservices for building cloud-based applications.
- You want to know if Java and Spring are relevant technologies for building microservice-based applications.
- You are interested in seeing what goes into deploying a microservice-based application to the cloud.

How this book is organized: A roadmap

This book consists of 12 chapters and 3 appendixes:

- Chapter 1 introduces you to why microservices architecture is an important and relevant approach to building applications, especially cloud-based applications.
- Chapter 2 walks you through the Spring Cloud technologies that we'll use and provides a guide on how to build cloud-native microservices following the twelve-factor application best practices. This chapter also walks you through how to build your first REST-based microservice using Spring Boot.
- Chapter 3 shows you how to look at your microservices through the eyes of an architect, application engineer, or DevOps engineer and provides a guide on how to implement some of the microservice best practices in your first REST-based microservice.

- Chapter 4 walks you through the container world, highlighting the main differences between containers and virtual machines (VMs). This chapter also shows you how to containerize your microservices using several Maven plugins and Docker commands.
- Chapter 5 introduces you to how to manage the configuration of your microservices using Spring Cloud Config. Spring Cloud Config helps guarantee that your service configurations are centralized in a single repository, versioned, and repeatable across all instances of your services.
- Chapter 6 introduces you to the service discovery routing pattern. You will learn how to use Spring Cloud and Netflix's Eureka service to abstract the location of your services away from the clients consuming them. You'll also learn how to implement client-side load balancing using the Spring Cloud LoadBalancer and a Netflix Feign client.
- Chapter 7 is about protecting the consumers of your microservices when one or more microservice instances are down or in a degraded state. This chapter demonstrates how to use Spring Cloud and Resilience4j to implement the circuit breaker, fallback, and bulkhead patterns.
- Chapter 8 covers the service gateway routing pattern. Using Spring Cloud Gateway, we build a single entry point to call all our microservices. We will demonstrate how to use the Spring Cloud Gateway filters to build policies that can be enforced against all services flowing through the service gateway.
- Chapter 9 covers how to implement service authentication and authorization using Keycloak. In this chapter, we cover some basic principles of OAuth2 and how to use Spring and Keycloak to protect your microservices architecture.
- Chapter 10 looks at how we can introduce asynchronous messaging into our microservices using Spring Cloud Stream and Apache Kafka. This chapter also shows you how to use Redis to cache lookups.
- Chapter 11 shows how to implement common logging patterns like log correlation, log aggregation, and tracing with Spring Cloud Sleuth, Zipkin, and the ELK Stack.
- Chapter 12 is the cornerstone project for this book. We take the services we have built throughout the book and deploy them to an Amazon Elastic Kubernetes Service (Amazon EKS). We also discuss how to automate the build and deployment of your microservices using tools like Jenkins.
- Appendix A shows additional microservices architecture best practices and explains the Richardson Maturity Model.
- Appendix B contains supplemental material on OAuth2. OAuth2 is an extremely flexible authentication model, and this chapter provides a brief overview of the different ways in which you can use OAuth2 to protect your application and its corresponding microservices.
- Appendix C covers how to monitor your Spring Boot microservices using several technologies such as Spring Boot Actuator, Micrometer, Prometheus, and Grafana.

In general, developers should read chapters 1, 2, and 3, which provide essential information about best practices and about implementing microservices using Spring Boot with Java 11. If you're a reader who's new to Docker, we highly recommend reading chapter 4 carefully because it briefly introduces all the Docker concepts used throughout the book.

The rest of the book discusses several microservice patterns, such as service discovery, distributed tracing, API Gateway, and more. The approach of this book is to read the chapters in order and follow the code examples for the chapters. But in case someone wants to skip the examples, they can download the code from the GitHub repository at **https://github.com/ihuaylupo/manning-smia**.

bout the code

This book contains code in almost every chapter, both in numbered listings and in line with normal text. In both cases, source code is formatted in a `fixed-width font like this` to separate it from ordinary text. All code examples are available in my GitHub repository at **https://github.com/ihuaylupo/manning-smia**.

Each chapter has a different folder in the repository. Note also that all the code in this book is built to run with Java 11 using Maven as the main build tool and Docker as the container tool. In the README.md file for each chapter, you can find the following information:

- A brief introduction to the chapter
- The tools required for the initial configuration
- A "how to use" section
- The build command for the examples
- The run command for the examples
- Contact and contributing information

One of the core concepts we followed throughout the entire book is that the code examples in each chapter should be able to run completely independent of any of the other chapters. What does this mean? You should be able to grab the code from chapter 10, for example, and run it without following the examples in the previous chapters. You'll see that for every service built in each chapter there is a corresponding Docker image. Each chapter uses Docker Compose to execute the Docker images in order to guarantee that you have a reproducible run-time environment for each chapter.

In many cases, the original source code has been reformatted; we've added line breaks and reworked indentation to accommodate the available page space in the book. In rare cases, even this was not enough, and listings include line-continuation markers (↪).

Additionally, comments in the source code have often been removed from the listings when the code is described in the text. Code annotations accompany many of the listings, highlighting important concepts.

veBook discussion forum

Purchase of *Spring Microservices in Action, Second Edition*, includes free access to a private web forum run by Manning Publications, where you can make comments about the book, ask technical questions, and receive help from the authors and from other users. To access the forum, go to **https://livebook.manning.com/book/spring-microservices-in-action-second-edition/discussion**. You can also learn more about Manning's forums and the rules of conduct at **https://livebook.manning.com/#!/discussion**.

Manning's commitment to our readers is to provide a venue where a meaningful dialogue between individual readers and between readers and authors can take place. It is not a commitment to any specific amount of participation on the part of the authors, whose contribution to the forum remains voluntary (and unpaid). We suggest you try asking them some challenging questions lest their interest stray! The forum and the archives of previous discussions will be accessible from the publisher's website as long as the book is in print.

about the authors

John Carnell is a software architect and leads the Developer Engagement team for Genesys Cloud. John spends the majority of his day teaching Genesys Cloud customers and internal developers how to deliver cloud-based contact center and telephony solutions and best practices for cloud-based development. He works hands-on building telephony-based microservices using the AWS platform. His day-to-day job is to design and build microservices across a number of technology platforms including Java, Clojure, and Go. John is a prolific speaker and writer. He regularly speaks at local user groups and was a regular speaker on "The No Fluff Just Stuff Software Symposium." Over the last 20 years, John has authored, coauthored, and functioned as a technical reviewer for a number of Java-based technology books and industry publications. John holds a BA from Marquette University and an MBA from the University of Wisconsin in Oshkosh. John is a passionate technologist who constantly explores new technologies and programming languages. When John is not speaking, writing, or coding, he lives in Cary, North Carolina, with his wife, Janet, his three children (Christopher, Agatha, and Jack), and yes, his dog, Vader.

Illary Huaylupo Sánchez is a software engineer who graduated from Cenfotec University and holds an MBA focused on IT management from the Latin American University of Science and Technology (ULACIT) in Costa Rica. Her knowledge of software development is quite extensive. She has experience working with Java and other programming languages such as Python, C#, Node.js, and with other technologies such as various databases, frameworks, cloud services, and more. Currently, Illary works as a Senior Software Engineer at Microsoft, San Jose, Costa Rica, where she spends most of her time researching and developing a variety of trendy and up-to-date projects. In her professional portfolio, we also find that she has 12 years of experience as an Oracle Certified Developer and has worked as a Senior Software Engineer in large companies such as IBM, Gorilla Logic, Cargill, and BAC Credomatic (a prestigious Latin American bank). Illary likes challenges and is always willing to learn new programming languages and new technologies. During her free time, she likes to play the bass guitar and spend time with her family and friends. Illary can be reached at **illaryhs@gmail.com**.

About the cover illustration

The figure on the cover of Spring Microservices in Action is captioned a "A Man from Croatia." This illustration is taken from a recent reprint of Balthasar Hacquet's Images and Descriptions of Southwestern and Eastern Wenda, Illyrians, and Slavs, published by the Ethnographic Museum in Split, Croatia, in 2008. Hacquet (1739–1815) was an Austrian physician and scientist who spent many years studying the botany, geology, and ethnography of many parts of the Austrian Empire, as well as the Veneto, the Julian Alps, and the western Balkans, inhabited in the past by peoples of the Illyrian tribes. Hand drawn illustrations accompany the many scientific papers and books that Hacquet published. The rich diversity of the drawings in Hacquet's publications speaks vividly of the uniqueness and individuality of the eastern Alpine and northwestern Balkan regions just 200 years ago.

This was a time when the dress codes of two villages separated by a few miles identified people uniquely as belonging to one or the other, and when members of a social class or trade could be easily distinguished by what they were wearing. Dress codes have changed since then and the diversity by region, so rich at the time, has faded away. It is now often hard to tell the inhabitant of one continent from another, and today the inhabitants of the picturesque towns and villages in the Slovenian Alps or Balkan coastal towns are not readily distinguishable from the residents of other parts of Europe.

We at Manning celebrate the inventiveness, the initiative, and the fun of the computer business with book covers based on costumes from two centuries ago, brought back to life by illustrations such as this one.

Previous chapter

< contents

Next chapter

1 Welcome to the cloud, Spring >